

Opinion

On the origin of memory neurons in the human hippocampus

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The hippocampus is essential for episodic memory, yet its coding mechanism remains debated. In humans, two main theories have been proposed: one suggests that concept neurons represent specific elements of an episode, while another posits a conjunctive code, where index neurons code the entire episode. Here, we integrate new findings of index neurons in humans and other animals with the concept-specific memory framework, proposing that concept neurons evolve from index neurons through overlapping memories. This process is supported by engram literature, which posits that neurons are allocated to a memory trace based on excitability and that reactivation induces excitability. By integrating these insights, we connect two historically disparate fields of neuroscience: engram research and human single neuron episodic memory research.

How the hippocampus codes episodic memories

Remember the last time you had coffee with a friend in your favorite coffee place. Loungy jazz music played in the background as a waiter attended another table. **Episodic memory** (see [Glossary](#)) [1,2] enables us to recall such past multimodal experiences with remarkable detail even after a single exposure. This process is fundamentally tied to the **hippocampus**. However, recent findings in humans are contradictory regarding how neurons in the hippocampus implement this episodic code. Current literature offers two possible explanations: one framework based on **concept neurons** coding specific elements within an episode [3–5] and an alternative based on **index neurons** coding the conjunction of these elements [6,7]. In this opinion, we first examine concept neurons in more depth, emphasizing their role as potential building blocks of episodic memory. We then explore index neurons, highlighting recent findings in humans and other species that provide strong support for a conjunctive coding framework. Next, we connect these two traditionally separate research streams by integrating insights from **engram** theory, specifically focusing on neuronal allocation via **excitability**. Building on this foundation, we propose a unified model in which index neurons gradually transform into concept neurons through overlapping memory traces.

Concept neurons: building blocks of episodic memory

Concept neurons are neurons in the human **medial temporal lobe** that fire in response to specific concepts in an all-or-none way [8–10]. These neurons are consistently found in humans and exhibit a high degree of multimodal invariance (i.e., they respond to an image of your friend or to their spoken name) and context invariance (i.e., they activate regardless of whether you see your friend at a movie, in a park, or in a café [10,11]). Curiously, the latency of their firing rate is much later than would be required by simple sensory processing and object recognition, which is an indication of their involvement in memory processing [11]. This lines up with the observation that most concept neurons are tuned to personally relevant concepts and depend on the subjective and conscious perception rather than on objective sensory properties [8,12]. In accordance with this, previous studies demonstrated that concept neurons increase their firing rate during memory retrieval

Highlights

The hippocampus plays a crucial role in episodic memory, with ongoing debate on how this is implemented at the level of single neurons.

Concept neurons encoding specific elements (e.g., a specific person) and index neurons that encode the conjunction of elements within an entire episode represent two primary theories in hippocampal coding.

Here, we reconcile these two frameworks by leveraging the discovery made in engram research that neurons are allocated to a memory trace based on excitability.

Based on this premise, we argue that, through the successive consolidation of episodes that share common elements, a subset of index neurons allocated to all these episodes would evolve into concept neurons tuned to this common element.

We suggest a focus on shared computational principles over the identification of discrete neural labels, as this approach allows a more productive discourse between researchers studying different species.

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when the image they are tuned to is part of the memory [8,13,14]. While concept neurons have been observed in both the hippocampus and amygdala, we focus here on the hippocampus, acknowledging that the roles of these regions may differ and require further investigation.

Concept neurons are not topographically organized, that is, spatially close concept neurons might code vastly different concepts [15]. This spatial organization allegedly benefits episodic memory processing as it allows association between any two concepts without connecting distant areas [4]. Quian Quiroga [3–5] proposed concept neurons as the building blocks of episodic memory formation and retrieval. Within this framework, a group of neurons collectively codes an episodic memory, with each neuron representing a specific element involved in that episode (Figure 1A). This means that when you meet your friend in your favorite café, the concurrent activation of one ensemble of concept neurons coding for your friend and another coding for the café represent the episode in the hippocampus. During memory retrieval, these assemblies then project back to the cortex, reinstating the sensory activity pattern first induced during the formation of the episode. A separate memory of the same friend in a different setting, such as a park, would, in turn, be represented by the simultaneous activity of the same ensemble coding for your friend and another ensemble representing the park [5].

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Indexing Theory: conjunctive code of episodic memories

An alternative framework, called the Indexing Theory, was proposed over three decades ago to explain hippocampal function during the encoding and retrieval of episodic memory in humans [6,7]. According to the Indexing Theory, during initial encoding, the various multimodal elements (the coffee shop, your friend, and the loungey jazz music) that comprise the episode instate a cortical activity pattern that is projected onto an ensemble of neurons in the hippocampus. Notably, the hippocampal ensemble does not code for any particular element within the episode, but

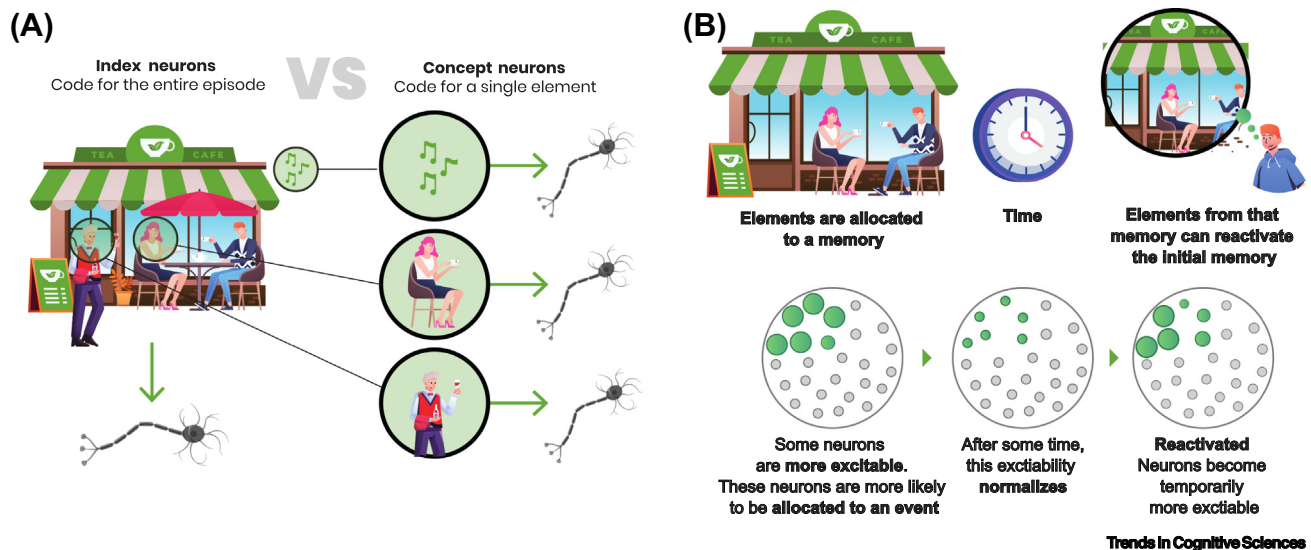


Figure 1. Two frameworks for hippocampal memory encoding and excitability as a driver of memory allocation. (A) Indexing Theory versus concept neurons. Left: according to the Indexing Theory, index neurons in the hippocampus conjunctively link the various elements of an episode (e.g., the loungey jazz music, your friend, and the waiter). Alternatively, under the concept neuron framework (right), each neuron codes for a distinct element of the episode, and their concurrent activity collectively represents the entire event. (B) Excitability decides memory allocation. Left: when an episodic memory (e.g., meeting a friend at a café) is formed, certain neurons are in a temporarily heightened state of excitability and, thus, are more likely to be allocated to that memory trace forming an engram (larger green circles). Center: over time, their excitability level gradually normalizes (shown by smaller green circles). Right: however, later exposure to any element from the original memory (e.g., seeing the café on a different occasion) can reactivate the memory trace. The reactivated neurons become temporarily more excitable again (larger green circles). This has a crucial role in the linking and updating of memories.

instead indexes (i.e., points toward) various coinciding elements (Figure 1A). This is also referred to as a **conjunctive code**. In other words, neurons code for the conjunction of the various elements within an episode rather than coding for the individual elements [6,7]. In keeping with the aforementioned example, one ensemble of index neurons codes for the entire episode at the coffee shop with your friend and the jazz music, instead of any individual element within.

Evidence for index neurons in humans and animals

In support of the Indexing Theory, hippocampal neurons in humans were recently reported that engage in conjunctive coding (called **episode specific neurons**) [16]. Specifically, these neurons were activated during the initial encoding of a unique episode (which involved associating three or two images) and demonstrated a reinstatement of their firing rate upon successful recollection of that same episode [16]. Reinstatement on the single neuron level was shown using both a **rate code** and a **temporal code**. These results were verified using several alternative reinstatement measures and changes in the hyperparameter space. Importantly, episode specific neurons were distinct from neurons that were tuned to specific concepts. Furthermore, these data suggested that episode specific neurons were predominantly observed in the hippocampus and likely excitatory pyramidal neurons [16]. For simplicity, we refer to these types of conjunctive neuron as index neurons throughout the remainder of this opinion. However, episode specific neurons are only putative index neurons. It has yet to be determined whether these neurons are involved in **pattern separation** (distinguishing similar episodes) and pattern completion (reinstating the full memory from partial cues) (Box 1). Additionally, it is unknown whether the activity pattern in the cortex that was present during the initial encoding of an episode is reinstated alongside the reactivation of the hippocampal index neurons during memory retrieval.

Further supporting evidence for index neurons in the hippocampus comes from studies on chickadees [17]. These food-caching birds provide a unique way to study the elements of episodic memories. Similar to episodic memories, caching sites must be learned in a single trial and contain

Box 1. Pattern completion and pattern separation in the hippocampus

Teyler and DiScenna's Indexing Theory offers a compelling framework for understanding how the brain encodes and retrieves episodic memories. This model highlights the interplay between the hippocampus and cortex in memory processes. After encoding an episodic memory, a partial input of the initial experience (e.g., you walk past the coffee shop and see the waiter attending a table) is sufficient to reactivate the entire ensemble of associated hippocampal index neurons, a process known as 'pattern completion'. These neurons then project back to the cortex, reinstating the entire experience [6,7].

Pattern separation refers to the complementary ability to distinguish between distinct episodes. Given that each experience is uniquely indexed, even highly overlapping cortical representations of similar episodes can be disambiguated in the hippocampus [6,7]. This hippocampal index allows a flexible way to quickly store the cortical representation of an episodic memory. Over time, the initially strengthened synaptic connections for unimportant memories either decay [102] or fall victim to interference [103]. Within this framework, the function of the hippocampus can be likened to a librarian: it directs to the relevant information within the library (the cortex) but does not have the knowledge itself. This implies that the hippocampus is content free because hippocampal neurons arbitrarily bind concurrent cortical activity irrespective of the semantic content they represent [7].

Concept neurons are neuronal assemblies in the human medial temporal lobe that respond selectively to specific concepts in an invariant way. Within this framework, pattern completion of episodic memories occurs through co-activation of these partially overlapping concept-specific neuronal assemblies. Consider walking past the familiar coffee shop. The concept neurons representing this location would activate, triggering a cascade of activation in associated concept neurons. This is possible because concept neuron assemblies exhibit small overlaps with other related assemblies. For instance, the neuronal group encoding the coffee shop might partially overlap with the ensemble representing the friend you met there, also activating that concept neuron ensemble. Research suggests that these overlaps between concept neuron assemblies can develop and refine over time through experience-dependent retuning of concept neurons [13]. However, this model presents a challenge: how does the hippocampus disambiguate highly similar memories, such as multiple visits to the same coffee shop under similar circumstances? The invariant nature of concept neurons alone cannot account for this level of specificity. To address this, it is necessary that additional neurons, such as time cells, work in conjunction with concept neurons to provide contextual information for distinguishing between similar episodes.

Glossary

Concept neurons: neurons in the human medial temporal lobe that respond to specific concepts, such as a person or object, playing a key role in memory and perception. They exhibit multimodal invariance, activating regardless of how the concept is presented (e.g., seeing a friend's picture or hearing their name) and context invariance, responding to the concept in any setting. For example, they might respond to a friend whether you see them in a café or a park.

Conjunctive code: way of representing information in the brain where neurons code for the combination of various elements within an episode (e.g., the location, people, and objects involved) rather than coding for each element individually.

Engram: physical trace of a memory in the brain, in the form of a group of neurons that have been changed or activated during the formation of a memory. Reactivation of engram neurons is sufficient and necessary to recall a memory.

Episode specific neurons: neurons that are active during the encoding of a specific episodic memory and reinstate their firing activity when the same episode is remembered. These neurons are thought to code for the entire episode as a whole (see 'Conjunctive code' earlier), rather than for specific elements within it.

Episodic memory: coined by Endel Tulving in 1972; the ability to encode and recall experiences containing a what, where, and when. Such memories are rich in detail, integrating information from multiple modalities, are encoded automatically without the need for repetitions, and can last a lifetime. By remembering these memories, it is as though we were mentally transported back in time, re-experiencing them anew.

Excitability: characteristic of neurons that describes how easily they fire an action potential in response to an incoming signal. Neurons with higher excitability are more likely to be involved in forming new memories.

Hippocampus: part of the brain located in the medial temporal lobe, crucial for forming, organizing, and storing memories, particularly episodic memories.

Index neurons: neurons that serve as pointers to the various elements of an

information about the ‘what’, ‘where’, and ‘when’. Observations show that, during caching, a small subset of pyramidal neurons in the hippocampus of the bird display significant increases in activity for a unique cache. At the same time, most other pyramidal neurons are suppressed, giving rise to a barcode-like pattern of activity unique to each food-caching event. Importantly, this specific pattern of activity is later reinstated during retrieval of the cached food. Notably, barcode activity was independent from place codes and distinct from other caching events at the same position [17].

Support for index neurons in computational models

Apart from empirical evidence, various models leverage the computational utility of index neurons. For instance, in a computational model of temporal attention, sparse index neurons (here termed ‘binding pool units’) were used to associate an item with its episodic context [18,19]. Through this index code, the authors captured order information, the repetition of identical or overlapping items, and enabled later retrieval of information in a neurally plausible way. Similarly, recent implementations of a biologically plausible modern Hopfield network integrated index neurons in the form of ‘memory neurons’ to allow the reconstruction of episodic and semantic memories [20,21] (for a transformer version of the Tolman–Eichenbaum Machine [22] see [23]).

Labeling of neuron types in the hippocampus: need for integration

The discovery of episode specific neurons inevitably added to the veritable *embarras de richesse* of neuron labels already defined in the human hippocampus. Apart from concept neurons and episode specific neurons, **place cells** have been discovered [24], as well as egocentric bearing cells [25], spatial target cells, serial position cells, heading-modulation cells [26], along with novelty cells, familiarity cells [27–29], and, in addition to, border cells, event cells [30] and time cells [31,32]. This raises the question: does the hippocampus indeed have a distinct neuron type for each of these cognitive functions? While these classifications help us understand specialized functions, this proliferation of neuron types might obscure the shared principles of how the hippocampus organizes information [33–36]. Therefore, here, we bridge the gap between two neuron types: index neurons and concept neurons. This requires delving into the rich literature on ‘engrams’, a type of memory neuron that has a history spanning over a century.

Engrams and neuronal allocation: the basis for memory traces

Historical background

An engram is a hypothetical construct of an ensemble of neurons that is roughly equivalent to a memory trace. The idea of an engram being critical for memory dates back to 1904, with the original coining of the term by Richard Semon [37]. Semon essentially proposed that neurons active at the time of an event are in some way changed (or undergo plasticity), which enables these neurons to be critical in storing the memory for that event. Subsequent reactivation of these ‘engram neurons’ by an external sensory cue or internal retrieval cue initiates retrieval of that memory. Rodent memory researchers have provided compelling evidence for the existence of engram ensembles and their importance in memory [38–47]. For instance, neurons that are active at the time of an event show a greater than chance likelihood of being reactivated when that specific memory is being retrieved [44,45,48–53]. Moreover, silencing neurons in an engram ensemble, but not a similar number of non-engram neurons, disrupts memory expression [45,54–56], while artificially activating neurons in an engram ensemble induces behavior consistent with memory, even in the absence of external memory retrieval cues [57–59]. Although early engram findings in rodents examined fear memories, where memory reactivation is typically measured via the expression of fear responses, recent work examined non-aversive memories in which memory is ‘read out’ as the mouse showing choice behavior [60]. Although the engram experiments in rodents do not exactly replicate the memory studies conducted in humans, we argue that the fundamental principles of engram formation and reactivation provide valuable insights into human memory processes.

episode, representing the conjunction of these elements. They do not code for specific details but rather link different aspects of an experience together.

Medial temporal lobe: brain region that includes the hippocampus and is important for memory formation.

Microwire: very thin wire used to record electrical activity from individual neurons in the brain.

Pattern separation: ability of the brain to distinguish between similar but distinct experiences or memories, allowing us to remember individual events separately.

Place cells: neurons in the hippocampus that become active when an animal (including humans) is in a specific location in its environment. These cells help create a mental map of our surroundings.

Rate code: type of neural coding where the firing rate of action potentials carries information.

Temporal code: type of neural coding where the timing of neuron firing carries information.

Neuronal competition and excitability

Computational and observational findings suggest that memories are sparsely encoded in relevant brain regions [48,61–64]. How then are neurons allocated to an engram ensemble? Research shows that, within a relevant brain region, such as the hippocampus, eligible neurons compete for allocation to a sparse engram ensemble [65]. Neurons with higher excitability during an experience are biased to win this competition and are allocated to the engram ensemble [44,47,55,66–68] (Figure 1B). Intrinsic neuronal excitability refers to the propensity of the neuron to fire an action potential in response to an input signal [69]. Excitability-based neuronal competition for engram allocation has been shown in a variety of brain regions [55,66], including the dorsal hippocampus [70,71] and in different types of memory [44,68,72,73]. These findings suggest that, at any given time, a small portion of eligible neurons in the hippocampus is ‘primed’ to become part of an engram (should an event occur) due to their increased excitability.

Memory linking and updating

Neuronal excitability is also important in linking or updating memories. For instance, two similar events can become linked if these events occur in close temporal proximity. Memory linking occurs because neurons allocated to the engram supporting the first event remain in an excited state for several hours after the event. Should a second similar event happen within this window of high engram neuron excitability then a portion of these neurons would also be allocated to the engram supporting the second event. This process has been termed ‘co-allocation’ and serves to link memories [70,73–78]. Moreover, when a memory is retrieved, the neurons in the engram ensemble supporting that memory are again reactivated and experience a transient increase in excitability [54,79,80] (Figure 1B). The reactivation of the engram ensemble of neurons allows new information to be incorporated into the memory [54] and for new memories to be layered on top [81]. Together, these findings indicate that the fundamental process of neuronal allocation allows information in the brain to be structured to facilitate the adaptive transformation of memories for individual experiences into organized conceptual knowledge.

The role of inhibitory neurons in engram formation

In previous work identifying episode specific neurons, evidence was found indicating that these neurons are excitatory, as suggested by their wider wave shapes [16]. While we primarily discuss index neurons and concept neurons in the context of excitatory neurons, inhibitory neurons have essential roles in modulating network activity and plasticity. Inhibitory interneurons, particularly GABAergic interneurons, have a crucial role in modulating this excitability during maturation [82]. There is also compelling evidence that inhibitory neurons can form engram-like structures, referred to as ‘inhibitory engrams’, which complement excitatory engrams [83]. These inhibitory representations arise through homeostatic potentiation of inhibition, counterbalancing excitatory neural assemblies and storing representations in a quiescent form until disinhibition occurs [83]. However, inhibitory neurons primarily balance or mask excitatory engrams rather than independently encoding unique concepts, suggesting that their role is fundamentally complementary, rather than identical, to excitatory engrams. While inhibitory neurons may not serve as index neurons or concept neurons directly, they are crucial in modulating network dynamics and influencing the processes by which index neurons and concept neurons emerge.

Integrating concept neurons and index neurons

Complementary roles in memory processing

It is conceivable that, in humans, index neurons work in complement with concept neurons during episodic memory processes (see Box 2 for how these two codes may map onto hippocampal pathways and subfields). Concept neurons might code the semantic aspects of an episode (i.e., the general concept of a coffee shop). As such, concept neurons might serve as a scaffold

Box 2. Hippocampal pathways and schemas in the hippocampus

There are two distinct codes in the hippocampus: a concept code formed through repeated encounters of similar elements, and an episodic-specific index that uniquely codes each event. How do these two codes map onto hippocampal subfields? A recent computational model [104] suggests a functional differentiation within hippocampal pathways. The monosynaptic pathway (MSP), connecting the entorhinal cortex directly to CA1, is critical for extracting semantic memory and statistical regularities, while the dentate gyrus and CA3, part of the trisynaptic pathway (TSP), represent event-specific information [104]. This functional division aligns with neurophysiological differences between the MSP and TSP: the MSP exhibits higher excitation and lower plasticity [105], properties that bias index neurons to develop into concept neurons representing frequently encountered concepts. By contrast, the TSP, connecting the entorhinal cortex to the dentate gyrus and CA3, shows more inhibition and higher plasticity [105], favoring the development of pattern-separated index neurons that add crucial event-specific information to episodes and distinguish similar events.

While schemas are traditionally associated with neocortical regions, the proposed role of concept neurons as interfaces or hubs within the hippocampus aligns with established findings and underscores complementary functions across these brain areas. While schemas in the neocortex form over extended periods, such as weeks [106], the hippocampus enables rapid integration of episodic and semantic information due to its higher synaptic plasticity [107]. Furthermore, concept neurons provide direct, efficient access to semantic concepts by reducing the need for extensive communication between the hippocampus and neocortex. Computational models support statistical learning in the hippocampus [104], showing that overlapping representations in the MSP [108] allow for generalization across environments by capturing commonalities. The learning rate of the MSP is well suited for rapid learning of statistical regularities over minutes and hours, acting as a filter before consolidation in the neocortex. A gradual shift toward lower learning rates from the TSP to the MSP and then to cortical areas [104,109] suggests that strict functional differentiation between hippocampus and neocortex are an oversimplification.

To conclude, we suggest that the hippocampus integrates both event-specific and concept-based coding to form cohesive episodic memories, with different pathways and subfields specializing in different aspects of memory processing. This hypothesis can be tested using ultra-high-resolution fMRI [110] or human single-unit recordings with improved localization techniques of microwire electrodes. Concept neurons, acting as an interface within the hippocampus, provide a scaffold for frequently encountered information and support rapid generalization, thus complementing the slower schema formation processes of the neocortex.

for memory by providing compressed representations for frequently encountered concepts, and giving these concepts prioritized access to memory [19]. This could explain why memory is heavily biased toward pre-existing schemas or semantic knowledge [84]. Within this line of thinking, concept neurons serve as an interface between semantically rich neocortical representations and a sparse, episodic specific code (Box 3). episode specific neurons may add crucial event-specific information to an episode that the highly general concept neurons cannot encode themselves (i.e., the type of music that was playing in the coffee shop) or allow separation of repeated visits to the same coffee shop with the same friend. This suggests complementary roles for concept neurons and episode specific neurons in memory processing. However, it neglects the open question of how concept neurons evolve in the first place. Later, we explain how this might happen, integrating the evidence presented earlier.

The evolution of concept neurons from index neurons

Imagine you meet your friend in a coffee shop. Based on findings in engram research, this coffee shop episode will initially be represented by a sparse ensemble of relatively more excitable neurons (Figure 2). Subsequently, these index neurons would act as a pointer to the entire event. A few days later, you meet up with the same person in a park and you remember the last time you met them in the coffee shop. This reactivates the index neurons coding the initial coffee shop episode. As discussed earlier in the section ‘Memory linking and updating’, engram literature suggests that neurons recently reactivated through retrieval, such as when recalling the coffee shop episode days later, experience a transient increase in excitability and, thus, are preferentially bound to a new episode [47,85]. This reactivation-induced excitability allows a subset of the neurons that initially encoded the coffee shop episode to be recruited into the engram supporting the park episode, even though the events are separated by days. This preferential allocation ensures consistent overlap across neurons coding experiences with shared elements. Of note, the overlap

Box 3. The role of inherent connectivity

Inherent connectivity is a fundamental factor influencing the functional role of a neuron within a network. Variations in structural connectivity can lead to substantial differences in neuronal behavior, as demonstrated by Kong and colleagues in their study on recurrent connectivity in the hippocampus [111]. Proximal CA3 (pCA3) neurons, with sparse recurrent connectivity, tend to support generalized contextual representations, while distal CA3 (dCA3) neurons, with denser recurrent connections, facilitate more discriminative representations [111]. Applying this to index neurons and concept neurons, it is plausible that neurons with different inherent connectivity patterns are predisposed to function as one or the other. Neurons with sparse recurrent connections may be more likely to become concept neurons due to their ability to integrate broader contextual information, whereas neurons with denser connectivity might function as index neurons by coding for specific episodic events through detailed integration of unique combinations of inputs. Seemingly contradicting this, Schapiro and colleagues [104] demonstrated in a computational model that sparse connectivity and high inhibition within the trisynaptic pathway were conducive to pattern separation of similar experiences. By contrast, the monosynaptic pathway, consisting of fully connected neurons with lower inhibition and a slower learning rate, led to the extraction of statistical regularities across experiences [104].

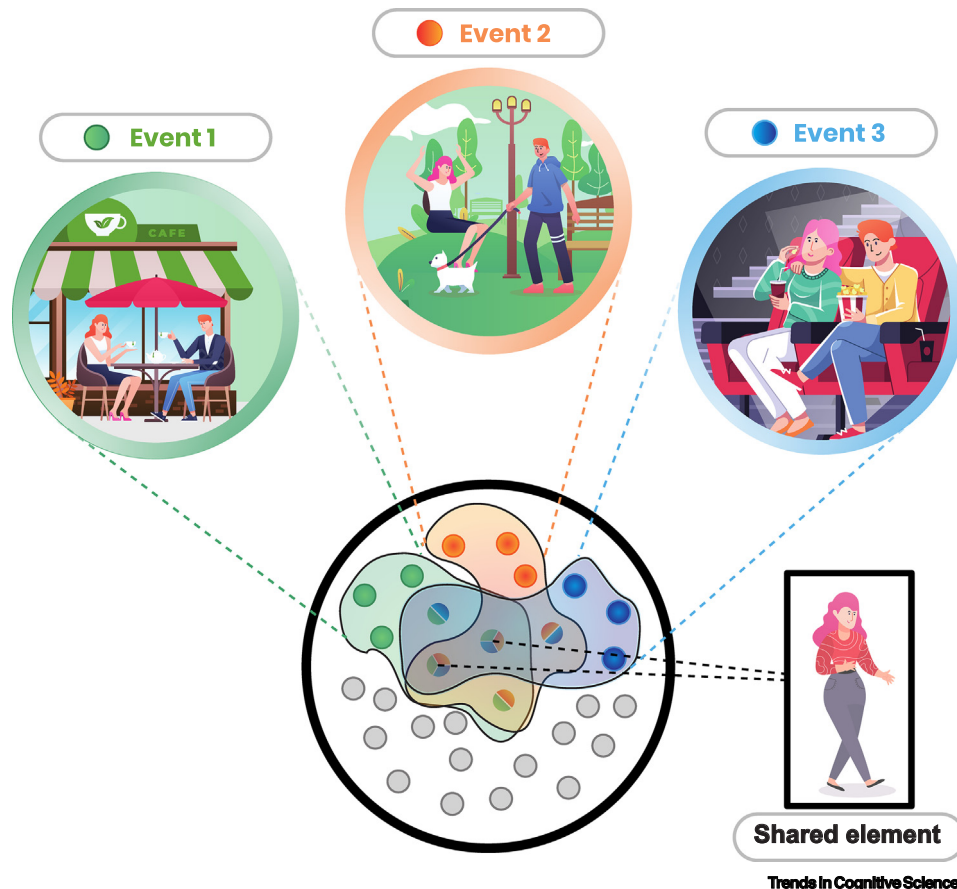
These contradictory results might be due to differences in the scale of analysis: Kong and colleagues [111] examined CA3 at a microscopic scale, whereas Schapiro and colleagues [104] focused on pathways at a mesoscopic scale. The non-uniform, often log-normal, distribution of synaptic connections and weights means that a small number of neurons receive a disproportionately large number of synaptic inputs or have stronger synaptic strengths [112]. Neurons with more or stronger synaptic inputs are more likely to be activated and reactivated, increasing their excitability and likelihood of being allocated to multiple memory traces, making them prime candidates for the transformation from index neurons into concept neurons.

In summary, inherent connectivity has a significant role in determining whether a neuron functions as an index neuron or is primed to transition into a concept neuron. Based on current literature, we predict that neurons within the monosynaptic pathway, neurons within the proximal CA3 subfield, and neurons with a larger number of synaptic connections are more likely to undergo the transformation from index neurons to concept neurons.

of neurons coding for each episode would be only partial so they can still be distinguished reliably. This balance of 'just enough' overlap allows index neurons to abstract common elements, while pattern separation ensures that each episode is encoded by a distinct set of neurons. Too much overlap can compromise this separation and cause interference between similar memories. A few days later when you watch a movie with the same friend, again a subset of assemblies coding for the previous two episodes will likely be bound to the third episode. The common element between all three episodes is your friend (Figure 2). It is conceivable that, over many such episodes, a proportion of the index neurons that initially coded the coffee shop episode would become 'semanticized' (i.e., develop a tuning for the friend). A concept neuron is born.

Throughout this process, neurons may exist on a continuum between episode-specific index neurons and highly abstract concept neurons. During these intermediate stages, neurons can code for multiple related concepts, especially if these concepts are associated through overlapping experiences. An illustrative example comes from observations of a neuron that responded to both actor Leslie Nielsen and images of airplanes due to their association through the movie *Airplane!* starring the actor [5]. Such neurons may represent an intermediate state, responding to multiple but related stimuli.

Neurons become less excitable as they age or are allocated to previous specific experiences, limiting their capacity for new allocations [54]. Newborn neurons exhibit heightened excitability and plasticity [82,86–88]. However, as they mature, their intrinsic excitability and firing rates decrease, leading to increased selectivity for specific inputs [89,90]. Therefore, while neurons can code for multiple related concepts, the allocation of concept neurons to completely unrelated episodes is unlikely, as their decreased excitability reduces their availability for encoding entirely new episodes. Consistent with this, when a new stimulus is associated with the tuned concept of a neuron, only 4% of concept neurons represent this newly associated stimulus over the long term [4,13].



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Figure 2. Overlapping memories transform a subset of index neurons into concept neurons. The top panel shows three different episodic events. The bottom panel illustrates how these events are allocated to neurons in the hippocampus, with colors indicating different episodes. According to findings in engram research, more excitable neurons are more likely to be allocated to an episodic memory and recently active neurons become more excitable. The first event shows you meeting a friend in a coffee shop (Event 1), which is allocated to the green neurons, which were more excitable at the time the event was experienced. The next day, during a park meeting with the same friend (Event 2), you remember the coffee shop episode from the previous day and the neurons representing the coffee shop become more excitable due to their recent activity. As a result, some of these neurons also start coding for the park episode. On the third day, meeting the same friend at the movies (Event 3) results in a further overlap, where a subset of neurons now codes for all three episodes. This process illustrates how index neurons, initially coding for specific episodes, may semanticize over repeated experiences to become concept neurons tuned to the shared element across all episodes: the friend.

Studies show that patients can develop concept neurons for experimenters they recently met [91], indicating that concept neurons can form over days rather than instantly, as seen with index neurons. This rapid formation in hospitals may be due to repetitive overlapping interactions with the same experimenters and heightened emotional experiences facilitating memory consolidation. In support of the latter, most concept neurons tuned to the experimenter were found in the amygdala [4,91]. Therefore, while concept neurons can appear to form quickly in emotionally charged, repetitive environments, this does not necessarily conflict with our hypothesis. Our proposal remains that, under typical circumstances, concept neurons emerge from the overlap of index neurons across multiple similar experiences.

In conclusion, we propose here the transformation of episode-specific index neurons into abstract concept neurons through the overlapping activation of neural assemblies across multiple related

experiences. This process allows neurons to gradually abstract common elements, such as a recurring friend in different settings, while maintaining distinct memories of each episode. As neurons mature and become less excitable, their capacity to encode entirely new, unrelated episodes diminishes, reinforcing their specialization. Emotional and repetitive experiences likely accelerate this process. For more details on how connectivity patterns within the hippocampus might influence whether a neuron remains an index neuron or eventually becomes a concept neuron, see Box 4.

Concluding remarks and future perspectives

Here, we integrated new findings on index neurons (neurons that code entire episodes conjunctively) with observations on element-specific concept neurons. To bridge these findings, we turned to a neurobiological mechanism identified in rodent engram research, which posits that neurons are first allocated to a memory trace based on excitability. The result is a unified framework where concept neurons evolve from index neurons, and both work in concert during episodic memory processing. This framework offers a new perspective on how the human hippocampus encodes and processes complex memories. By integrating these insights, we connect two historically disparate fields of neuroscience: engram research and human single-neuron episodic memory research. We conclude by providing future directions and falsifiable hypotheses that will shape our understanding of the human memory system and inform neurotechnology interventions to enhance or restore memory function (see [Outstanding questions](#)).

Our proposed model of the transformation of index neurons into concept neurons likely unfolds over extended periods and potentially involves reactivations during sleep [92–95]. Therefore, to fully understand this phenomenon, it is crucial for future studies to implement a longitudinal design, recording neuronal activity continuously over multiple days.

Based on our framework we make the following predictions: (i) **microwire** stimulation could provide causal evidence for an excitability-based memory code in humans [96,97]. Stimulating neurons near a microwire should increase the probability that these neurons are allocated to an episode, shedding light on the allocation mechanism of index neurons and potentially increasing the yield of index neurons per patient. Given the described lack of topographical organization in the hippocampus, stimulating neurons may also increase the excitability of neurons already allocated to unrelated episodes. However, although neuronal excitability is typically downregulated several hours after allocation, this downregulation is not absolute. Microwire stimulation may further elevate the excitability of previously allocated neurons beyond the recruitment threshold [98]. While this may introduce some noise by involving neurons from unrelated episodes, it does not undermine the transformation from index neurons to concept neurons, as the process can otherwise continue unaffected; (ii) while increasing excitability via microwire stimulation should enhance the probability of neuron allocation to an episode, decreasing excitability should reduce this likelihood; (iii) by stimulating during multiple episodes that share a common element, for example, encountering a friend in different settings, neurons may develop invariant responses to that common element, transforming into concept neurons. (iv) conversely, inhibiting these neurons or preventing their reactivation during the encoding of related memories would cause the second event to be allocated to a different group of neurons and could prevent the formation of concept neurons. (v) reactivating previously assigned neurons through electrical stimulation could increase the probability of memory retrieval or even induce retrieval without a cue; (vi) reactivation through retrieval is expected to increase the excitability of allocated neurons, which should show a larger propensity of firing an action potential as a response to electrical stimulation; and (vii) the transformation from index neurons to concept neurons should be accompanied by specific changes, such as shifts in firing patterns or connectivity, which could be identified using longitudinal studies. Overall, these aforementioned experiments could provide valuable insights into the mechanisms of memory formation and retrieval.

Outstanding questions

Are episode specific neurons equivalent to index neurons as proposed by Teyler and DiScenna? Key index neuron features remain untested, including whether: (i) partial inputs suffice to reactivate the episode specific neuron ensemble (pattern completion); (ii) similar episodes activate distinct episode specific neuron ensembles (pattern completion); (iii) firing patterns of episode specific neurons remain consistent over time and multiple retrievals; and (iv) cortical activation patterns precede hippocampal index activity during encoding and the hippocampal index projects back to the cortex during successful retrieval.

Do index neurons undergo excitability-based competition for memory allocation, similar to engram cells in rodents? Investigating whether neuronal excitability predicts index neuron involvement in episodic memory could provide insights into shared principles of memory formation across species.

Does the development of concept neurons reflect a process of semanticization that comes at the cost of detailed memory retrieval of distinct episodes? As index neurons semanticize into concept neurons over multiple retrievals, does this result in a loss of detailed memory for individual episodes? Do related memories become more similar over time, potentially reducing the ability to recall distinct details?

Are concept neurons and index neurons differentially involved in the subjective experience of memory? Does the strength of reactivation of concept neurons versus index neurons during memory retrieval correlate with different aspects of the subjective experience of memory, such as vividness, emotional intensity, and the richness of details?

How many episodic experiences are necessary for index neurons to 'semanticize' into concept neurons? What role does sleep or memory replay have in this transition of consolidating overlapping episodes into a coherent concept?

How does the hippocampus integrate episode specific neurons and concept neurons to form a cohesive episodic memory?

Linking engram allocation theory and Indexing Theory has important implications for the development of a neural memory prosthesis. Consider a device that excites a small population of neurons in the hippocampus during a specific event [96]. This event could later be retrieved by reactivating the same neuronal population on demand. Conversely, inhibiting this population could suppress the memory. This concept outlines a potential blueprint for a memory prosthesis capable of tagging and modulating memories as needed. Such a device offers a promising approach to alleviate symptoms in memory disorders, such as Alzheimer's disease or post-traumatic stress disorder (PTSD).

The yield of neurons using currently available electrodes is about a dozen per microwire bundle. By comparison, electrodes developed more recently yield hundreds of well-localized neurons [99–101]. This difference in neuron yield cannot always be compensated for by recording more participants or sessions. For instance, recording hundreds of neurons in one patient would enable the analysis of between-cell interactions, such as the detection of assemblies of index neurons, which is not feasible when recording only a few neurons. Having access to multiple neurons that are allocated to the same episode would create the opportunity to ask more nuanced questions. If a cue does not initiate memory retrieval, do some neurons reinstate their firing activity, although the activation is not sufficient to activate the entire ensemble? Is there a relationship between the strength of ensemble reinstatement and the detail of episodic retrieval? It is conceivable that the answers will differ depending on the hippocampal subfield. Given that we lack sufficient coverage to record multiple neurons of one ensemble that codes an episode, we cannot currently investigate these questions.

In neuroscience research, the choice of experimental paradigms is often driven by practical considerations related to the species being studied and the constraints of recording methods. In rodent studies, where direct communication about subjective experiences is not possible, researchers focus on behaviors that can be readily quantified, such as spatial navigation, making the study of place-coding neurons attractive. In human recordings, safety and practical limitations often necessitate that patients remain relatively immobile; thus, visual or auditory stimuli are preferred over movement-related studies. Presenting a large number of images and identifying neurons that respond selectively becomes a fruitful research endeavor. However, this difference in paradigms often makes research between species hard to compare. The labels we place on responsive neurons appear diametral to one another, because the experimental input is fundamentally different. While directly translating experimental designs between species is one approach, it can be difficult to implement effectively. Instead of attempting to find matching experimental designs between species that do not translate well, we suggest looking for common computational principles. For example, engram cells in rodents and episode specific neurons in humans both exhibit conjunctive coding properties. Highlighting their shared conjunctive nature allows us to leverage the extensive animal research and propose excitability as a common neural allocation principle. This similarity allows us to hypothesize that index neurons in humans could form the basis of concept neurons.

We recognize that determining computational functions is challenging. However, we believe that focusing on computational principles, where feasible, allows a more fruitful discourse between researchers studying different species.

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Declaration of interests

The authors declare no competing interests.

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